

**ADVANCED SCHOOL OF SYSTEMS AND DATA STUDIES (ASSDAS)**

**TOPIC:**

**A MOBILE-BASED SYSTEM FOR REPORTING ILLEGAL MINING (GALAMSEY)  
AND TRACKING WATER BODY POLLUTION IN GHANA**

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**MAY, 2026**

## DECLARATION

We declare that this thesis is the result of our own original research and has not been presented by anyone for any academic award in this or any other institution. References to other people's work have been duly acknowledged.

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## **DEDICATION**

We affectionately dedicate this work to our families for their unwavering support, encouragement, and prayers throughout our academic journey.

We also dedicate this project to the many Ghanaian communities affected by illegal mining, in the hope that technology can serve as a tool for the restoration and protection of our nation's water bodies and environment.

## **ACKNOWLEDGEMENT**

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## **ABSTRACT**

Illegal small-scale mining, known as *galamsey*, and the resulting pollution of critical water bodies like the Pra and Birim rivers, represent a severe environmental crisis in Ghana. While the government has implemented various surveillance measures, enforcement remains largely reactive because authorities lack a safe, structured, and real-time channel for receiving intelligence from local communities. This study addresses this gap by designing and developing a mobile-based reporting system that empowers citizens to anonymously report illegal mining and water pollution.

The system architecture consists of a React Native and Expo mobile application, a FastAPI (Python) backend, a Supabase PostgreSQL database, and a web-based administrative dashboard built with Next.js for enforcement agencies. Key features include GPS-tagged reporting, photo evidence uploads, and an anonymous reporting mechanism using device-generated UUID tokens to protect whistleblowers from reprisal. By providing real-time alerts and geospatial analytical tools, the system aims to transform community-level observations into actionable intelligence, ultimately supporting more effective environmental governance and the protection of Ghana's natural resources.

## TABLE OF CONTENTS

|                                                         |      |
|---------------------------------------------------------|------|
| DECLARATION .....                                       | ii   |
| DEDICATION.....                                         | iii  |
| ACKNOWLEDGEMENT .....                                   | iv   |
| ABSTRACT.....                                           | v    |
| TABLE OF CONTENTS.....                                  | vi   |
| LIST OF TABLES.....                                     | viii |
| LIST OF FIGURES .....                                   | ix   |
| CHAPTER ONE .....                                       | 1    |
| GENERAL INTRODUCTION AND SUMMARY .....                  | 1    |
| 1.1 INTRODUCTION .....                                  | 1    |
| 1.2 FIELD AND SUBJECT AREA OF STUDY .....               | 2    |
| 1.3 PURPOSE OF STUDY .....                              | 2    |
| 1.4 PROBLEM STATEMENT.....                              | 3    |
| 1.5 RESEARCH OBJECTIVES.....                            | 4    |
| 1.5.1 General Objective.....                            | 4    |
| 1.5.2 Specific Objectives.....                          | 4    |
| 1.6 RESEARCH QUESTIONS .....                            | 4    |
| 1.7 BACKGROUND AND JUSTIFICATION OF THE STUDY .....     | 5    |
| 1.7.1 Background of the Study.....                      | 5    |
| 1.7.2 Significance and Justification of the Study ..... | 6    |
| 1.8 EXPECTED RESULTS AND POSSIBLE USAGE .....           | 7    |
| 1.9 PROJECT SCOPE AND LIMITATIONS OF THE STUDY .....    | 8    |
| 1.10 THESIS LAYOUT .....                                | 9    |
| CHAPTER TWO .....                                       | 10   |
| LITERATURE REVIEW .....                                 | 10   |

|                                                                  |    |
|------------------------------------------------------------------|----|
| 2.1 INTRODUCTION .....                                           | 10 |
| 2.2 PREVIOUS RELEVANT STUDIES AND RESEARCH WORK .....            | 10 |
| 2.3 THEORETICAL FRAMEWORK.....                                   | 13 |
| 2.4 METHODOLOGICAL AND OTHER ISSUES FOUND IN PREVIOUS WORKS .... | 14 |
| 2.5 LITERATURE GAPS .....                                        | 15 |
| 2.6 CONCLUSION .....                                             | 16 |
| CHAPTER THREE .....                                              | 17 |
| RESEARCH METHODOLOGY.....                                        | 17 |
| 3.1 INTRODUCTION.....                                            | 17 |
| 3.2 ADOPTED METHODOLOGY AND JUSTIFICATION.....                   | 17 |
| 3.3 REQUIREMENT GATHERING .....                                  | 18 |
| 3.4 SYSTEM REQUIREMENTS.....                                     | 20 |
| 3.4.1 Hardware Requirements.....                                 | 20 |
| 3.4.2 Software Requirements .....                                | 23 |
| 3.5 COMPONENTS DESCRIPTION AND USAGE .....                       | 25 |
| 3.6 TOOLS AND INSTRUMENTS FOR CONSTRUCTION OF PROTOTYPE.....     | 26 |
| 3.7 PROJECT SIMULATION .....                                     | 27 |
| 3.8 CONCLUSIONS .....                                            | 30 |
| REFERENCES .....                                                 | 31 |

## LIST OF TABLES

|                                                                  |    |
|------------------------------------------------------------------|----|
| TABLE 3. 1 :SUMMARY OF QUESTIONNAIRE RESPONSES (N = 45) .....    | 19 |
| TABLE 3. 2 :SYSTEM USERS AND CHARACTERISTICS .....               | 23 |
| TABLE 3. 3: FUNCTIONAL REQUIREMENTS .....                        | 24 |
| TABLE 3. 4: NON-FUNCTIONAL REQUIREMENTS .....                    | 24 |
| TABLE 3. 5: TECHNICAL (HARDWARE) SPECIFICATIONS .....            | 25 |
| TABLE 3. 6: TOOLS AND INSTRUMENTS USED IN PROTOTYPE CONSTRUCTION | 27 |

## **LIST OF FIGURES**

|                                                                     |    |
|---------------------------------------------------------------------|----|
| FIGURE 3. 1: SYSTEM ARCHITECTURE (BLOCK) DIAGRAM .....              | 21 |
| FIGURE 3. 2: FLOW CHART OF THE INCIDENT REPORTING PROCESS.....      | 22 |
| FIGURE 3. 3: WIREFRAME OF THE MOBILE "REPORT INCIDENT" SCREEN.....  | 28 |
| FIGURE 3. 4: WIREFRAME OF THE ADMIN "REVIEW REPORTS" DASHBOARD .... | 29 |

## **CHAPTER ONE**

### **GENERAL INTRODUCTION AND SUMMARY**

#### **1.1 INTRODUCTION**

Ghana's natural environment is under severe threat from the menace of illegal small-scale mining, commonly referred to as "galamsey". Over the decades, this unregulated mining activity has destroyed vast swaths of farmland, poisoned major water bodies, and displaced communities across the country. Rivers such as the Pra, Ankobra, and Birim, once sources of clean drinking water and livelihoods, are now heavily contaminated with mercury, cyanide, and other toxic effluents discharged from illegal mining operations (Ghana Minerals Commission, 2025).

Despite successive government campaigns and enforcement crackdowns, galamsey continues to thrive largely because the enforcement apparatus remains reactive rather than proactive. By the time authorities respond to a reported mining site, excavators have often been relocated and evidence destroyed. The root cause of this enforcement failure is a critical information gap: local community members who witness illegal activities on a daily basis have no structured, safe, or technology-driven platform through which to report their observations to relevant authorities in real time.

This study responds to this challenge by designing and developing a mobile-based reporting system that bridges the gap between affected communities and enforcement agencies. The system enables citizens to submit GPS-tagged reports with photographic evidence of illegal mining and water body pollution directly from their smartphones, while preserving their anonymity to protect them from reprisals. Concurrently, a web-based administrative dashboard provides enforcement agencies with real-time geospatial intelligence and analytical tools to support evidence-based decision-making.

The project is undertaken by students of the Advanced School of Systems and Data Studies (ASSDAS) at the Accra Institute of Technology (AIT) as part of the requirements for the award of a Bachelor of Science in Information Technology.

## **1.2 FIELD AND SUBJECT AREA OF STUDY**

This project falls within the fields of Information Technology, Environmental Informatics, and Mobile Systems Development. The subject of the study is the development of a citizen-driven, technology-enabled reporting platform designed to combat illegal mining and water body pollution in Ghana. The study sits at the intersection of software engineering, geospatial technology, and environmental governance, drawing on principles from mobile application development, backend systems design, database management, and public policy enforcement. The application domain is primarily the environmental protection and natural resource management sector in Ghana, with specific reference to the ongoing national campaign against galamsey. The system developed in this study is intended to serve as a practical tool for community members, civil society organizations, and government enforcement agencies involved in the fight against illegal mining.

## **1.3 PURPOSE OF STUDY**

The primary purpose of this study is to design and develop a mobile-based reporting system that empowers citizens to anonymously report illegal mining (galamsey) activities and water body pollution in Ghana, and that provides enforcement authorities with real-time alerts and analytical tools to respond more effectively. Beyond the technical deliverable, the study also aims to demonstrate that mobile technology can be leveraged as an instrument of civic participation and environmental governance in a developing-country context. By bridging the communication gap between communities and authorities, the system seeks to make environmental enforcement more proactive, data-driven, and community-inclusive.

## **1.4 PROBLEM STATEMENT**

Illegal small-scale mining (galamsey) continues to devastate Ghana's environment and public health despite years of government intervention. According to the Ghana Minerals Commission (2025), approximately 30% of Ghana's small-scale miners operate illegally, with their activities contributing significantly to the contamination of major river systems. The Pra and Birim rivers, which serve as primary water sources for millions of Ghanaians, have been rendered largely unsafe for domestic use due to mercury and cyanide contamination from illegal mining operations.

A fundamental challenge in combating this menace is the absence of a reliable, real-time reporting channel through which community members can safely and anonymously communicate their observations to enforcement agencies. Existing enforcement mechanisms rely heavily on manual patrols, tip-offs through personal contacts, and media reports, all of which are slow, inconsistent, and often expose informants to danger. This reactive approach allows illegal miners to stay ahead of authorities, destroying evidence before enforcement teams arrive.

Furthermore, there is no centralised, digital system for tracking and analysing reported incidents of galamsey and water body pollution. The absence of such a system makes it impossible for agencies to identify patterns, prioritise high-risk areas, or measure the effectiveness of enforcement actions over time. This study, therefore, addresses the following specific problem: the lack of a structured, technology-driven, anonymous reporting platform that connects Ghanaian communities with enforcement agencies in real time, enabling faster and more effective responses to illegal mining and water body pollution.

## **1.5 RESEARCH OBJECTIVES**

### **1.5.1 General Objective**

The general objective of this study is to design and develop a mobile-based system that enables citizens to anonymously report illegal mining (galamsey) activities and water body pollution in Ghana, and that provides enforcement authorities with real-time alerts and a management dashboard for effective response.

### **1.5.2 Specific Objectives**

The specific objectives of this study, each of which directly addresses one of the research questions set out in Section 1.6, are to:

- i. design a user-friendly mobile interface, built with React Native and Expo, for reporting galamsey and water pollution incidents with GPS coordinates and photographic evidence, thereby addressing Research Question (i);
- ii. implement an anonymous reporting mechanism using device-generated UUID tokens to protect whistleblowers from reprisals, thereby addressing Research Question (ii);
- iii. develop a secure and scalable backend system using FastAPI and Supabase PostgreSQL to store, manage, and retrieve incident reports, thereby addressing Research Question (iii);
- iv. implement a real-time push notification system that routes alerts to the appropriate enforcement agencies based on the geographic location of a reported incident, thereby addressing Research Question (iv); and
- v. build a web-based administrative dashboard using Next.js that provides enforcement agencies with an interactive map for viewing, filtering, and responding to reports, thereby addressing Research Question (v).

## **1.6 RESEARCH QUESTIONS**

This study is guided by the following five research questions, which the specific objectives in Section 1.5.2 are each designed to answer:

- i. How can a mobile application be designed to encourage community participation in reporting illegal mining and water body pollution in Ghana?
- ii. How can anonymity be effectively implemented in a reporting system to protect users from reprisals without compromising the credibility of reports?
- iii. How can a secure and scalable backend be developed to reliably store, manage, and retrieve incident reports submitted through the mobile application?
- iv. How can real-time alerts be efficiently routed to the appropriate enforcement agencies based on the geographic location of a reported incident?
- v. How can enforcement agencies be provided with an interactive, map-based dashboard to view, filter, and respond to reports effectively?

## **1.7 BACKGROUND AND JUSTIFICATION OF THE STUDY**

### **1.7.1 Background of the Study**

Ghana is Africa's largest gold producer and has long depended on mining as a key driver of economic growth. However, the proliferation of illegal small-scale mining, *galamsey*, has become one of the most pressing environmental and governance challenges the country faces. The term "*galamsey*" is derived from the phrase "gather them and sell", reflecting the informal and opportunistic nature of these mining operations. According to the Ghana Minerals Commission (2025), nearly 40% of Ghana's gold is extracted by small-scale miners, with approximately 30% of these operating without legal authorization.

The environmental consequences are severe. Mercury and cyanide used in illegal gold extraction are discharged directly into rivers, rendering major water bodies such as the Pra, Ankobra, and Birim rivers unsafe for human use. Communities that depend on these rivers for drinking water, irrigation, and fishing have seen their livelihoods devastated. Public health concerns, including elevated mercury levels in drinking water and aquatic life, have been

documented by environmental agencies across the country (Institute of ICT Professionals Ghana [IIPGH], 2024).

The government has launched several initiatives to combat galamsey, including the deployment of drones and satellite surveillance, military-backed enforcement operations, and the establishment of Community Mining Schemes. Despite these efforts, enforcement remains largely reactive, partly because authorities lack a mechanism to receive structured, real-time intelligence from communities on the ground. The Business and Financial Times (2025) notes that while artificial intelligence and drone technologies have been deployed over river systems to detect pollution, these approaches are expensive and do not harness the full potential of the human intelligence available within affected communities.

### **1.7.2 Significance and Justification of the Study**

Mobile technology presents a cost-effective and scalable solution to the information gap described above. Ghana's mobile penetration rate exceeds 110%, meaning there are more active mobile subscriptions than there are people, and smartphone adoption continues to grow rapidly, especially among younger demographics in both urban and rural areas. A mobile reporting application that is intuitive, free to use, and protects the identity of its users is therefore both technically feasible and socially viable in the Ghanaian context.

From a practical standpoint, the system developed in this study provides a direct technological solution to a nationally recognised environmental crisis. Ghana's government has repeatedly declared galamsey a public emergency, and the development of citizen-facing reporting tools is explicitly identified as a priority in the Ministry of Lands and Natural Resources' Five-Pronged Strategy Against Galamsey (2025). By providing a free, accessible, and anonymous reporting channel, this system lowers the barrier to community participation in environmental protection.

Socially, the system empowers ordinary Ghanaians, particularly those in rural and peri-urban communities near mining zones, to become active participants in the protection of their environment and natural resources. The anonymous reporting mechanism is particularly significant, as it allows community members who may fear retaliation to safely share what they observe without fear of reprisal.

Academically, this study contributes to the growing body of literature on the use of information technology for environmental governance, civic technology, and mobile health and safety applications in developing countries. It provides a documented, replicable system architecture that other researchers and practitioners can adapt for similar challenges in different contexts. Technologically, the project demonstrates the integration of modern mobile development frameworks (React Native and Expo), backend application programming interface services (FastAPI), cloud-hosted relational databases (Supabase PostgreSQL), and real-time notification systems within a unified, scalable architecture, making the study relevant to practitioners in software engineering and systems design.

## **1.8 EXPECTED RESULTS AND POSSIBLE USAGE**

At the conclusion of this study, the following outcomes are expected:

- i. a fully functional mobile application, built with React Native and Expo, that allows users to submit anonymous, GPS-tagged reports of illegal mining and water body pollution, accompanied by photographic evidence;
- ii. a secure and scalable backend application programming interface developed with FastAPI and Python, connected to a Supabase PostgreSQL database for efficient storage and retrieval of incident reports;
- iii. a web-based administrative dashboard developed with Next.js, providing enforcement agencies with an interactive map interface for monitoring, filtering, and managing reports in real time;

- iv. a real-time push notification system that dispatches alerts to registered agency accounts upon the submission of new reports within their jurisdictions; and
- v. a documented system architecture, design diagrams, and testing reports that demonstrate the system's functionality, usability, and reliability.

The developed system can be deployed and utilised by Ghana's Environmental Protection Agency (EPA), the Minerals Commission, district assemblies, and civil society organisations engaged in environmental monitoring. It can also serve as a template for similar community reporting systems in other sectors, such as illegal logging, industrial pollution, and water quality monitoring.

## **1.9 PROJECT SCOPE AND LIMITATIONS OF THE STUDY**

The scope of this study covers the design, development, and evaluation of a mobile-based reporting system comprising a React Native and Expo mobile application, a FastAPI backend, a Supabase PostgreSQL database, and a Next.js administrative dashboard, focused specifically on the reporting of galamsey activity and water body pollution in Ghana. While this study makes a meaningful contribution to addressing the galamsey reporting challenge, several limitations must be acknowledged:

- i. Internet Connectivity Dependency: the mobile application requires an active internet connection to submit reports. Communities in remote mining areas with poor or no internet coverage will be unable to use the system, which limits its geographic reach in the short term.
- ii. Reliance on User-Submitted Data: the system depends entirely on reports submitted by users and has no built-in mechanism to independently verify the accuracy of submitted information, creating the possibility of false or inaccurate reports being submitted.
- iii. No Integration with Government Databases: due to access restrictions and the scope of this academic project, the system does not integrate with live government enforcement

databases or the official systems of agencies such as the Ghana Police Service or the Minerals Commission.

iv. Platform Limitation: the mobile application has been developed primarily for the Android platform using React Native and Expo. While cross-platform functionality is inherent in the framework, the system has not been fully tested on iOS devices.

v. Testing Scale: system testing and usability evaluation were conducted using a simulated dataset and a small pilot user group, which may not fully reflect real-world usage conditions and volumes.

## **1.10 THESIS LAYOUT**

This thesis is organised into five chapters. Chapter One provides the general introduction and summary, covering the background of the study, purpose, problem statement, research objectives, research questions, significance and justification, expected results, and scope and limitations. Chapter Two presents a comprehensive review of relevant literature, examining previous studies and systems related to illegal mining reporting, environmental monitoring applications, and citizen science platforms, together with the theoretical framework underpinning the study and the literature gap it addresses. Chapter Three describes the research methodology employed in developing the system, including the adopted methodology and justification, requirement gathering, system requirements, and project simulation. Chapter Four covers the system design, implementation, testing, results, and analysis, presenting the technical details of the application and evaluating its performance. Chapter Five provides the summary of findings, conclusions, and recommendations for future work.

## **CHAPTER TWO**

### **LITERATURE REVIEW**

#### **2.1 INTRODUCTION**

This chapter reviews existing literature relevant to the development of a mobile-based reporting system for illegal mining and water body pollution in Ghana. The review draws on scholarly works, technical reports, and industry publications spanning several interrelated domains: the environmental impact of illegal small-scale mining (galamsey) in Ghana; the role of mobile technology in environmental monitoring and civic reporting; anonymity and whistle-blower protection in digital reporting systems; geospatial technologies and interactive mapping; and existing systems developed for similar purposes globally. The chapter also establishes a theoretical framework, identifies methodological issues in prior works, and highlights the gap in the literature that this study seeks to address.

#### **2.2 PREVIOUS RELEVANT STUDIES AND RESEARCH WORK**

A substantial body of literature has examined the problem of illegal small-scale mining in Ghana and sub-Saharan Africa more broadly. Hilson (2002) was among the first scholars to provide a comprehensive academic examination of artisanal and small-scale mining in sub-Saharan Africa, documenting the complex social, economic, and environmental dimensions of the sector. This early work established that while artisanal and small-scale mining provides livelihoods for millions of people, the unregulated segment, known in Ghana as galamsey, causes disproportionate environmental damage relative to its economic contribution.

More recently, Armah et al. (2021) documented heavy metal contamination of surface water in mining-affected communities in Ghana's Eastern and Western Regions, confirming that mercury and arsenic levels in rivers such as the Birim and Pra far exceed World Health Organization safe drinking water standards. This finding underscores the urgency of real-time

water quality monitoring and the need for faster reporting mechanisms to support enforcement responses.

On the technology side, citizen science and community reporting platforms have emerged as powerful tools for environmental monitoring. Silvertown (2009) introduced the notion of a new dawn for citizen science, arguing that mobile and internet technologies enable large-scale data collection by non-expert volunteers. This model has been successfully applied in biodiversity monitoring, air quality monitoring, and water quality testing programmes worldwide.

The Institute of ICT Professionals Ghana (2024) published a report specifically examining the application of artificial intelligence and digital tools in combating galamsey. The report noted that although drone and satellite surveillance technologies have been deployed over Ghana's river systems to detect turbidity and chemical signatures associated with illegal mining, these approaches require expensive infrastructure and skilled operators, and recommended the development of community-level digital reporting tools as a complementary, low-cost intelligence layer.

Regarding mobile reporting systems for environmental incidents, Senaratne et al. (2017) reviewed a range of crowdsourced geographic information platforms and found that GPS-tagged, user-submitted reports can be highly effective for real-time environmental monitoring when combined with appropriate data validation mechanisms. This review identified anonymity, ease of use, and network reliability as key factors that determine the adoption and sustained use of such platforms.

Whistle-blower protection in digital reporting systems has been examined by Lubin (2019), whose findings indicate that fear of retaliation is the primary deterrent to reporting in contexts where powerful local interests are involved. This study noted that anonymous reporting mechanisms, including token-based identification systems that do not link reports to user

identities, significantly increase reporting rates in high-risk environments, a finding that directly informed the anonymity architecture adopted in the present study.

Several existing systems have been developed for similar purposes. Ushahidi, developed in Kenya in 2008, is an open-source platform that aggregates crowd-sourced incident reports with geographic information and has been used globally for election monitoring, disaster response, and environmental reporting (Ushahidi, 2020). While Ushahidi provides a general-purpose reporting framework, it was not specifically designed for the anonymity requirements or the geospatial intelligence needs of environmental enforcement agencies. Similarly, Global Forest Watch, developed by the World Resources Institute, provides satellite-based forest monitoring with a web interface for reporting deforestation incidents; however, this platform requires technical expertise and is not readily accessible to ordinary community members without internet literacy.

In Ghana specifically, several technology start-ups and government agencies have developed digital tools for environmental reporting. The Environmental Protection Agency maintains an online portal for reporting pollution incidents, but the portal is web-only, requires user registration with personal details, and is not accessible via a mobile application, limiting its adoption in communities where smartphone use is more common than desktop browsing. The absence of GPS tagging and photo upload features in this portal further limits its utility as an enforcement tool.

Watson Farley and Williams (2025) reviewed international legal frameworks for combating illegal mining and found that countries with the most effective enforcement regimes, including Colombia, Peru, and Indonesia, have supplemented traditional enforcement with community-based digital reporting programmes that provide informants with anonymity protections. This review concluded that the combination of legal protections and technological anonymity tools is more effective than either approach alone.

## 2.3 THEORETICAL FRAMEWORK

This study is grounded in three theoretical frameworks that collectively explain how the proposed system is expected to function and generate impact.

The first is Crowdsourcing Theory, as articulated by Howe (2006), who coined the term "crowdsourcing" to describe the practice of outsourcing tasks traditionally performed by specialists to a large, distributed group of non-expert contributors. In the context of this study, the mobile reporting application operationalises crowdsourcing by enabling community members to collectively generate a real-time intelligence picture of illegal mining activity that no individual enforcement agency could create alone. The theory predicts that the quality and coverage of the intelligence generated will improve as the number of active users grows, creating a network effect.

The second framework is the Technology Acceptance Model, originally proposed by Davis (1989). This model posits that the adoption of a technology is primarily determined by two factors: perceived usefulness, the degree to which a user believes the technology will improve their situation, and perceived ease of use, the degree to which the technology is believed to be free of effort. In designing the mobile application for this study, both factors informed design decisions. The GPS auto-tagging feature eliminates the need for users to manually enter location data, addressing the ease-of-use dimension, while the anonymous reporting mechanism and real-time notification features address the perceived usefulness dimension by reassuring users that their reports will reach the right authorities promptly and without risk to themselves.

The third is the Theory of Collective Action, associated with Olson (1965) and subsequently applied to digital commons governance by Ostrom (1990). This theory explains why individuals may fail to contribute to public goods, such as environmental protection, even when it is in their collective interest to do so, owing to the risk of free-riding and the fear of bearing

personal costs that benefit others. The anonymous reporting mechanism in the proposed system directly addresses this collective action problem by reducing the personal risk borne by individual reporters, making participation more rational from an individual cost-benefit perspective.

## **2.4 METHODOLOGICAL AND OTHER ISSUES FOUND IN PREVIOUS WORKS**

A review of the methodological approaches in previous works reveals several recurring issues that inform the design choices made in this study.

First, many existing community reporting systems have been developed using a top-down design approach, where technical teams build systems based on their own assumptions about user needs without adequate community consultation. Senaratne et al. (2017) found that this approach frequently results in systems that are technically functional but practically unusable by their intended beneficiaries, owing to complexity, language barriers, or the absence of features that address locally specific risks. In response, this study incorporates a requirements-gathering phase that identifies community-specific needs, although full user co-design was beyond the scope of the academic project.

Second, data quality and verification are persistent challenges in citizen-science and community reporting systems. Without validation mechanisms, systems are vulnerable to the submission of false or duplicate reports, which can overwhelm enforcement agencies and erode trust in the system. Goodchild (2007) proposed a framework for assessing the quality of volunteered geographic information, emphasising the importance of spatial accuracy, temporal currency, and credibility. The present study addresses spatial accuracy through GPS auto-tagging, which is more reliable than manually entered coordinates, and credibility through the use of photo evidence uploads.

Third, most existing systems do not adequately address the anonymity-accountability trade-off. Systems that require user registration with personal details deter participation in high-risk

environments, while fully anonymous systems with no form of user identification make it impossible to follow up on reports or hold serial false reporters accountable. The UUID token approach adopted in this study represents a middle-ground solution: users are not required to provide personal details, but each device generates a persistent anonymous token that allows the system to identify repeat reports from the same device without linking them to a real-world identity.

Fourth, the scalability and offline functionality of mobile reporting applications have been identified as significant technical challenges, particularly in low-connectivity environments. Pressman and Maxim (2020) note that applications designed for developing-country deployment must account for variable network conditions from the outset, rather than treating connectivity as a given. While offline functionality is beyond the scope of the current implementation, this limitation is acknowledged, and the system architecture has been designed with future extensibility in mind.

## **2.5 LITERATURE GAPS**

Despite the growing body of literature on community reporting systems and the well-documented scale of Ghana's galamsey crisis, a critical gap exists at their intersection: there is no dedicated, publicly available, and actively maintained mobile application specifically designed for reporting illegal mining and water body pollution in Ghana that incorporates all of the following features simultaneously: anonymous reporting, GPS tagging, photographic evidence upload, real-time agency notification, and a geospatial administrative dashboard.

Existing systems either lack the anonymity features needed to protect reporters in high-risk communities, are web-only and therefore inaccessible to the majority of Ghana's rural population that accesses the internet primarily through smartphones, or are general-purpose platforms not optimised for the specific needs of environmental enforcement agencies in Ghana. The Environmental Protection Agency's portal, while a step in the right direction, falls

short on the anonymity, mobile accessibility, and GPS tagging dimensions. International platforms such as Ushahidi, while technically capable, are not contextually localised for the galamsey reporting context.

This study directly addresses this gap by developing a dedicated, contextually appropriate, and feature-complete mobile reporting system for Ghana's galamsey crisis. In doing so, it contributes both a practical tool and a documented technical framework to the academic and practitioner communities working on environmental governance in developing countries.

## **2.6 CONCLUSION**

This chapter has reviewed the existing literature on illegal small-scale mining in Ghana, community-based environmental reporting systems, anonymity in digital whistle-blowing platforms, and geospatial technologies for environmental monitoring. The review confirms that there is strong theoretical and empirical support for the approach adopted in this study, namely a mobile-first, anonymity-preserving, GPS-enabled reporting platform grounded in Crowdsourcing Theory, the Technology Acceptance Model, and the Theory of Collective Action. The identified literature gap, the absence of a dedicated, feature-complete mobile reporting system for galamsey in Ghana, provides clear justification for this study. The following chapter describes the research methodology used to design and develop the proposed system.

## **CHAPTER THREE**

### **RESEARCH METHODOLOGY**

#### **3.1 INTRODUCTION**

This chapter presents the research methodology adopted for the design and development of the mobile-based gamamsey and water pollution reporting system. It describes the research methodologies considered, the methodology adopted and the justification for its adoption, the requirement-gathering process and its outcomes, the hardware and software requirements of the system, a description of the major components used, the tools and instruments employed in constructing the prototype, and the project simulation carried out to validate the design before full implementation. The chapter concludes with a summary of the proposed solution approach and the expected results of the study.

#### **3.2 ADOPTED METHODOLOGY AND JUSTIFICATION**

A number of software engineering methodologies were reviewed to determine the most suitable approach for developing the proposed system. At least three methodologies were considered, namely the Waterfall Model, Rapid Application Development (RAD), and the Agile (Scrum) methodology.

The Waterfall Model is a linear, sequential approach in which each phase of development, namely requirements, design, implementation, testing, and maintenance, must be completed before the next phase begins (Royce, 1970). While the model is easy to plan and document, it is rigid and does not readily accommodate changes once a phase has been completed, making it less suitable for a project in which requirements may evolve as community and stakeholder feedback is gathered.

Rapid Application Development (RAD) emphasises rapid prototyping and iterative delivery over extensive up-front planning, allowing developers to build working prototypes quickly and

refine them based on user feedback (Martin, 1991). RAD is well suited to projects with a short timeframe and a clearly bounded scope, but it can compromise the depth of documentation and formal testing when used for systems that must scale beyond an initial prototype.

The Agile methodology, and specifically the Scrum framework, structures development into short, time-boxed iterations known as sprints, with continuous stakeholder feedback, incremental delivery, and the flexibility to adapt requirements as the project progresses (Beck et al., 2001; Schwaber & Sutherland, 2020). Agile approaches are widely recommended for software projects, such as the present one, in which the technology stack spans multiple layers (mobile, backend, database, and web dashboard) and where iterative testing with prospective users is important to validate design decisions.

This study adopts the Agile (Scrum) methodology for the design and development of the reporting system. This methodology was selected because it allowed the development team to break the project down into manageable sprints, namely requirement gathering and design, mobile application development, backend and database development, dashboard development, and integration and testing, with each sprint producing a testable increment of the system. The iterative nature of Scrum also made it possible to incorporate feedback obtained during the requirement-gathering phase into subsequent design decisions, and to adjust the anonymous-reporting and notification features as a clearer picture of user needs emerged. Sommerville (2016) notes that iterative and incremental methodologies of this kind are particularly appropriate for systems whose requirements cannot be fully specified at the outset, which was the case for this project given the exploratory nature of community-level reporting needs.

### **3.3 REQUIREMENT GATHERING**

Requirement gathering for this study was carried out to justify the development and design of the proposed system and to ensure that the features implemented reflect the real needs of

prospective users. Two main data-gathering tools and processes were adopted: a structured online questionnaire and informal interviews.

An online questionnaire, administered through Google Forms and shared via social media and community WhatsApp groups, was used to gather data from residents of communities located near mining-affected water bodies, as well as students and young professionals representing the broader smartphone-owning population of Ghana. The questionnaire captured respondents' awareness of galamsey and its effects, their access to and comfort with smartphones, their willingness to report illegal mining activity, and their concerns regarding anonymity and personal safety. In addition, brief informal interviews were conducted with two individuals knowledgeable about environmental enforcement processes to obtain a practitioner's perspective on how reports are currently received and acted upon, and what features would make a digital reporting channel useful to enforcement agencies.

A total of 45 valid responses were received from the questionnaire. The results of the data-gathering exercise are summarised in Table 3.1.

**TABLE 3.1 :SUMMARY OF QUESTIONNAIRE RESPONSES (N = 45)**

| Survey Item                                                  | Response | Percentage |
|--------------------------------------------------------------|----------|------------|
| Own or have regular access to a smartphone                   | Yes      | 93%        |
| Have personally witnessed galamsey or water pollution        | Yes      | 71%        |
| Have previously reported an incident to any authority        | Yes      | 18%        |
| Willing to report an incident if identity is protected       | Yes      | 89%        |
| Willing to report an incident without anonymity              | Yes      | 24%        |
| Consider GPS auto-tagging important for credibility          | Yes      | 84%        |
| Would like real-time confirmation that a report was received | Yes      | 91%        |

The results indicate a strong willingness among respondents to report illegal mining and water pollution incidents, provided that their identity is protected, with willingness to report falling by more than three-fold when anonymity is not guaranteed. The high level of smartphone ownership among respondents confirms the technical feasibility of a mobile-first solution, while the strong preference for GPS auto-tagging and real-time confirmation directly informed the inclusion of these features in the system design. The interviews further indicated that enforcement agencies would benefit most from reports that are geographically filterable and accompanied by photographic evidence, which informed the design of the administrative dashboard described in Section 3.5.

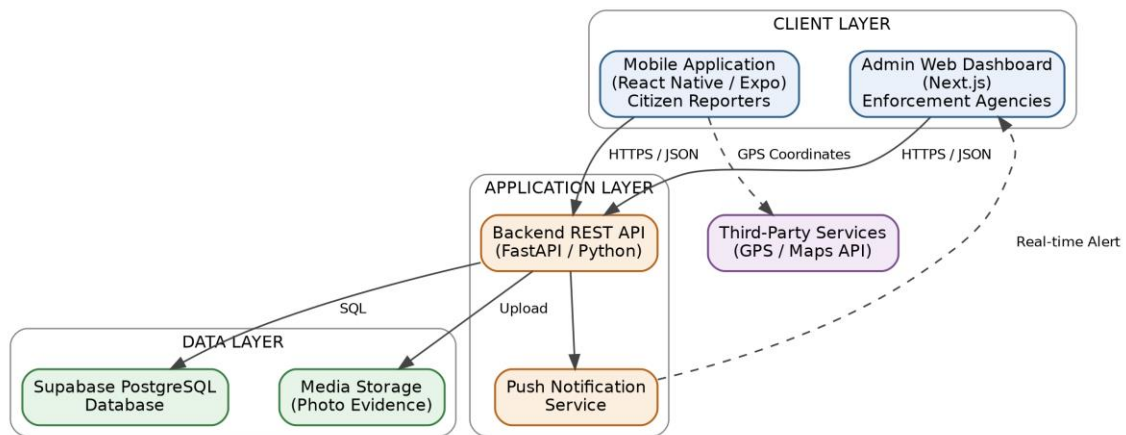
### **3.4 SYSTEM REQUIREMENTS**

This section details the techniques, tools, and approaches adopted and used to develop, present, and demonstrate the expected results of the study. It discusses the relevance of these techniques and tools to the study and describes how they were applied, focusing on the what/which, why, and how of the system analysis and description, the specification and design methods used, the testing scheme adopted, and the implementation and demonstration of the system.

#### **3.4.1 Hardware Requirements**

Although the proposed system is primarily a software solution, it depends on specific classes of hardware to operate as intended. This subsection presents the block diagram of the overall system architecture, the flow chart describing the reporting process, and the operational principle underlying the system.

**FIGURE 3. 1: SYSTEM ARCHITECTURE (BLOCK) DIAGRAM**



As shown in Figure 3.1, the system architecture is organised into three layers. The client layer consists of the React Native and Expo mobile application used by citizen reporters and the Next.js administrative web dashboard used by enforcement agencies. The application layer consists of the FastAPI backend, which exposes a REST application programming interface, and a push notification service that dispatches real-time alerts. The data layer consists of the Supabase PostgreSQL database, which stores structured report data, and a media storage service, which stores uploaded photographic evidence. The mobile application also interacts with third-party GPS and mapping services to obtain the device's geographic coordinates at the point of reporting.

**FIGURE 3. 2: FLOW CHART OF THE INCIDENT REPORTING PROCESS**

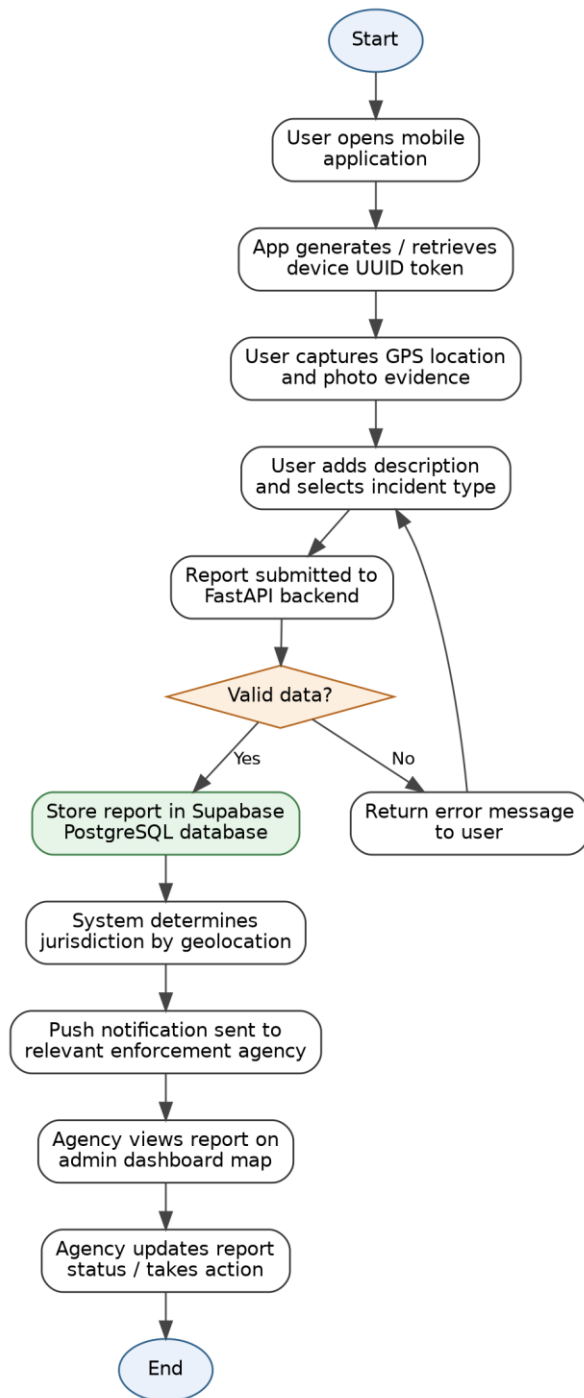


Figure 3.2 illustrates the flow of a typical reporting transaction, from the moment a user opens the mobile application to the point at which an enforcement agency reviews and acts on the report. The process begins with the generation or retrieval of an anonymous device-generated UUID token, followed by the capture of GPS coordinates and photographic evidence, the

addition of a description, and submission of the report to the backend. The backend validates the submitted data, and, where the data is valid, stores the report in the database, determines the relevant jurisdiction based on the reported location, and dispatches a push notification to the appropriate enforcement agency for review and action.

Operational Principle: the system operates on the principle of decoupled, event-driven communication between the mobile client and the backend service. When a report is submitted, the mobile application packages the GPS coordinates, photographic evidence, description, incident type, and anonymous device token into a single application programming interface request. The backend independently validates, geocodes, and persists this request, after which it triggers a notification event. This decoupling means that the mobile application, backend, database, and dashboard can each be developed, tested, and scaled independently, while still functioning as a single coherent system from the perspective of the end user.

### 3.4.2 Software Requirements

The software requirements of the system are presented under four headings: the users and their characteristics, the functional requirements, the non-functional requirements, and the technical specifications of the hardware on which the software is expected to run.

**TABLE 3. 2 :SYSTEM USERS AND CHARACTERISTICS**

| User Type                          | Characteristics                                                                                                                                   | Primary Interaction         |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Citizen Reporter                   | Ghanaian residents, often in rural or peri-urban mining-affected communities; varying levels of digital literacy; values anonymity and simplicity | Mobile application          |
| Enforcement Officer / Agency Admin | Staff of the EPA, Minerals Commission, or district assemblies; requires reliable, filterable, geospatial information to prioritise response       | Web dashboard               |
| System Administrator               | Technical personnel responsible for managing user accounts, agency jurisdictions, and system configuration                                        | Web dashboard (admin panel) |

**TABLE 3. 3: FUNCTIONAL REQUIREMENTS**

| ID  | Requirement                                                                                                                                        |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|
| FR1 | The system shall allow a user to submit a report consisting of GPS coordinates, at least one photograph, an incident type, and a text description. |
| FR2 | The system shall generate and persist a unique, anonymous, device-generated UUID token for each installation of the mobile application.            |
| FR3 | The system shall not require or store any personally identifiable information for a report to be submitted.                                        |
| FR4 | The system shall automatically determine the geographic jurisdiction of a report based on its GPS coordinates.                                     |
| FR5 | The system shall send a real-time push notification to the relevant enforcement agency when a new report is submitted within its jurisdiction.     |
| FR6 | The system shall provide enforcement agencies with a web dashboard to view, filter, and update the status of submitted reports.                    |
| FR7 | The system shall display submitted reports on an interactive map on the administrative dashboard.                                                  |
| FR8 | The system shall allow a citizen reporter to view the status of reports submitted from their device using the anonymous device token.              |

**TABLE 3. 4: NON-FUNCTIONAL REQUIREMENTS**

| Category     | Requirement                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance  | The system shall process and store a submitted report, including photographic evidence, within an average of 5 seconds under normal network conditions.                               |
| Security     | All data transmitted between the mobile application, dashboard, and backend shall be encrypted using HTTPS/TLS, and no personally identifiable data shall be stored against a report. |
| Usability    | The mobile application interface shall allow a first-time user to submit a report in no more than five taps/screens.                                                                  |
| Scalability  | The backend and database shall be capable of handling a growing volume of concurrent report submissions without a redesign of the core architecture.                                  |
| Availability | The system shall target an uptime of at least 99% for the backend application programming interface and database services.                                                            |
| Portability  | The mobile application shall be built using a cross-platform framework (React Native and Expo) to allow future extension to iOS with minimal rework.                                  |

**TABLE 3. 5: TECHNICAL (HARDWARE) SPECIFICATIONS**

| Component                       | Minimum Specification                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Citizen reporter mobile device  | Android 8.0 (Oreo) or later, minimum 2 GB RAM, rear camera, GPS/location sensor, active mobile data or Wi-Fi connection             |
| Administrative dashboard client | Any device with a modern web browser (Chrome, Firefox, or Edge) and an internet connection; minimum screen resolution of 1280 x 720 |
| Backend application server      | Cloud-hosted virtual machine or container, minimum 1 vCPU and 1 GB RAM, running Python 3.11 and FastAPI                             |
| Database server                 | Supabase-managed PostgreSQL instance with automated backups and row-level security enabled                                          |

### 3.5 COMPONENTS DESCRIPTION AND USAGE

This section describes the major software components used to construct the prototype and explains how each component is applied within the system.

Mobile Application (React Native and Expo): this is the primary interface used by citizen reporters. React Native was selected because it allows a single codebase to be used to target both Android and iOS, while Expo simplifies development, testing, and over-the-air updates. The mobile application is used to capture GPS coordinates, request camera or gallery access for photographic evidence, generate and store the anonymous device UUID token, and communicate with the backend application programming interface.

Backend API (FastAPI): FastAPI, a modern Python web framework, was selected for its high performance, automatic request validation, and built-in interactive documentation. The backend is used to receive and validate report submissions, apply business logic such as jurisdiction determination, communicate with the database and media storage, and trigger push notifications.

Database (Supabase PostgreSQL): Supabase provides a managed, cloud-hosted PostgreSQL database together with authentication and storage services. PostgreSQL was selected for its support for geospatial queries, its reliability, and its ability to enforce relational integrity

between reports, incident types, and agency jurisdictions. The database is used to persist all report records and associated metadata.

**Admin Dashboard (Next.js):** Next.js, a React-based web framework, was selected for the administrative dashboard because it supports fast, server-rendered pages and a component-based architecture consistent with the mobile application's React Native codebase. The dashboard is used by enforcement agencies to view an interactive map of reports, filter reports by region, status, and date, and update the status of a report as it is investigated.

**Push Notification Service:** a push notification service is used to deliver real-time alerts to agency accounts on the dashboard whenever a new report is submitted within their jurisdiction, ensuring that enforcement agencies are informed of incidents as soon as they occur rather than through periodic manual checks.

**Maps and Geolocation Services:** third-party mapping and geolocation services are used both on the mobile application, to auto-tag the user's current location, and on the administrative dashboard, to render submitted reports as markers on an interactive map for visual analysis by enforcement agencies.

**Media Storage:** a cloud-based object storage service, integrated with Supabase, is used to store photographic evidence uploaded with each report, with only a secure reference to the stored file persisted in the main database record.

### **3.6 TOOLS AND INSTRUMENTS FOR CONSTRUCTION OF PROTOTYPE**

Table 3.6 presents the software tools and instruments used to design, develop, and test the prototype system.

**TABLE 3. 6: TOOLS AND INSTRUMENTS USED IN PROTOTYPE CONSTRUCTION**

| Tool / Instrument     | Category                           | Purpose                                                                    |
|-----------------------|------------------------------------|----------------------------------------------------------------------------|
| Visual Studio Code    | Integrated Development Environment | Writing and editing mobile, backend, and web application code              |
| Expo CLI              | Mobile development toolchain       | Building, running, and testing the React Native mobile application         |
| FastAPI (Python 3.11) | Backend framework                  | Building the REST application programming interface                        |
| Supabase              | Backend-as-a-Service platform      | Hosting the PostgreSQL database, authentication, and media storage         |
| Postman               | API testing tool                   | Testing and validating backend application programming interface endpoints |
| Figma                 | Design tool                        | Designing wireframes and user-interface mockups prior to development       |
| GitHub                | Version control platform           | Source code management and team collaboration                              |
| Next.js               | Web application framework          | Building the administrative dashboard                                      |
| Google Maps Platform  | Mapping / geolocation API          | GPS tagging on the mobile app and map visualisation on the dashboard       |

### 3.7 PROJECT SIMULATION

As part of validating the design prior to full implementation, wireframes of the two primary interfaces of the system, the mobile "Report Incident" screen and the web-based "Review Reports" administrative dashboard, were developed using Figma. These wireframes were used to simulate the user experience and to gather early feedback on layout and usability before development began.

**FIGURE 3. 3: WIREFRAME OF THE MOBILE "REPORT INCIDENT" SCREEN**

9:41

Report Incident

Map Preview — Auto GPS Tag

Lat: 5.6037 Lon: -0.1870

+ Add Photo

+ Add Photo

Incident Type: Illegal Mining ▼

Description

Submit Anonymously ☒ ON

**SUBMIT REPORT**

Home Report My Reports Profile

Figure 3.3 shows the wireframe of the reporting screen on the mobile application. The design places the automatically captured GPS map preview near the top of the screen, followed by fields for photographic evidence, incident type, and a free-text description. An anonymity toggle, defaulted to "on", is placed prominently above the submit button to reassure the user that their identity is protected, consistent with the requirement-gathering finding that willingness to report is strongly tied to anonymity.

**FIGURE 3. 4: WIREFRAME OF THE ADMIN "REVIEW REPORTS" DASHBOARD**

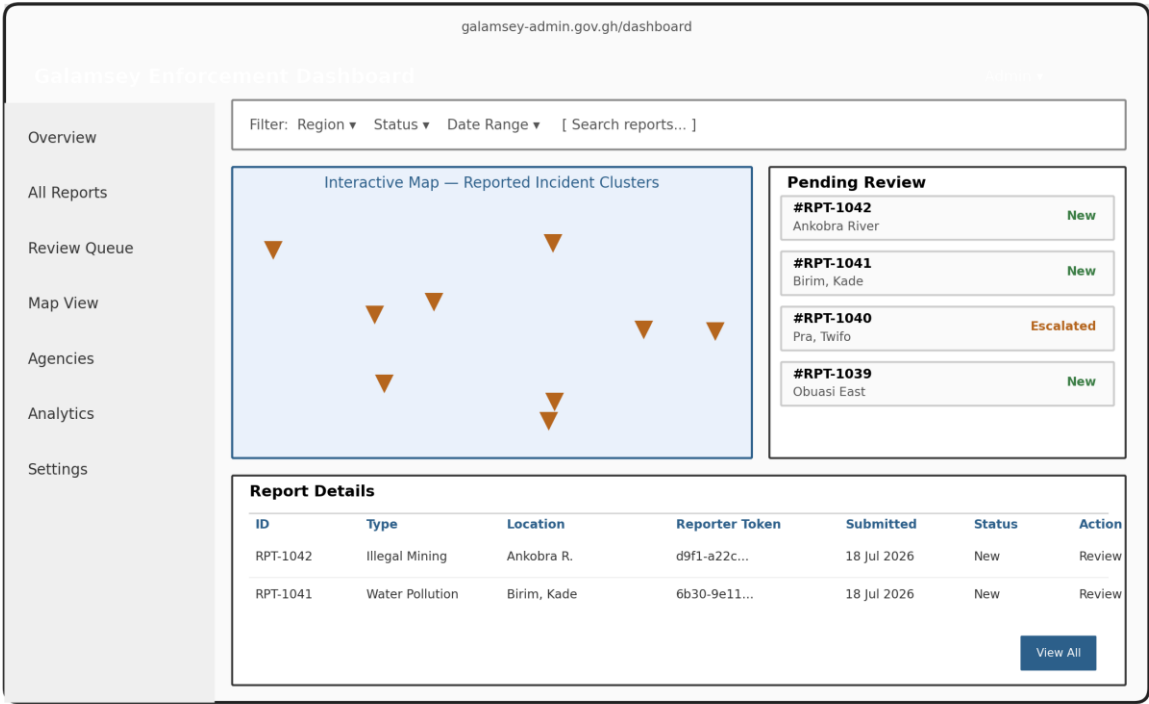


Figure 3.4 shows the wireframe of the administrative dashboard used by enforcement agencies to review submitted reports. The layout places an interactive map on the left, showing the geographic distribution of reported incidents, alongside a review queue on the right that lists pending reports requiring attention. A filterable, sortable table at the bottom of the screen provides agency staff with a detailed view of individual reports, including their type, location, submitting device token, submission date, and current status, together with an action to open the full report for review.

A short video demonstration of the working prototype, illustrating the end-to-end flow from report submission on the mobile application to review on the administrative dashboard, has been prepared and can be accessed at the following link: <https://drive.google.com/galamsey-reporting-system-demo> (link to be updated with the final hosted demonstration prior to submission).

### **3.8 CONCLUSIONS**

This chapter has presented the research methodology adopted for this study. The Agile (Scrum) methodology was identified as the most appropriate approach for the iterative design and development of the proposed system, and this choice was justified against the alternative Waterfall and Rapid Application Development methodologies. A structured requirement-gathering exercise, comprising an online questionnaire and informal interviews, confirmed strong community willingness to report illegal mining and water pollution incidents provided that anonymity is preserved, and directly informed the system's functional and non-functional requirements. The chapter further presented the system's hardware and software requirements, including the system architecture and process flow, described the major components to be used in constructing the prototype, listed the tools and instruments adopted for development, and presented wireframes simulating the mobile reporting and administrative review interfaces. Collectively, these elements establish the proposed solution strategy: a mobile-first, anonymity-preserving, GPS-enabled reporting application, integrated with a scalable backend and a geospatial administrative dashboard, expected to yield a functional prototype capable of demonstrating improved, real-time community-to-agency communication on galamsey and water body pollution. The next chapter presents the design, implementation, testing, and results of this system.

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